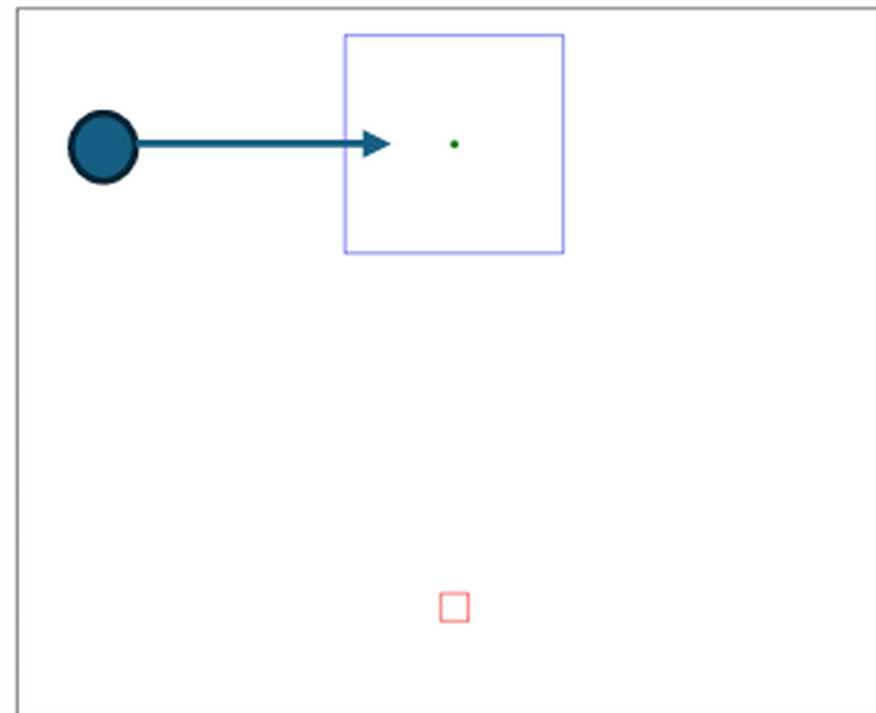


Compact Representations of Human Interception Movements

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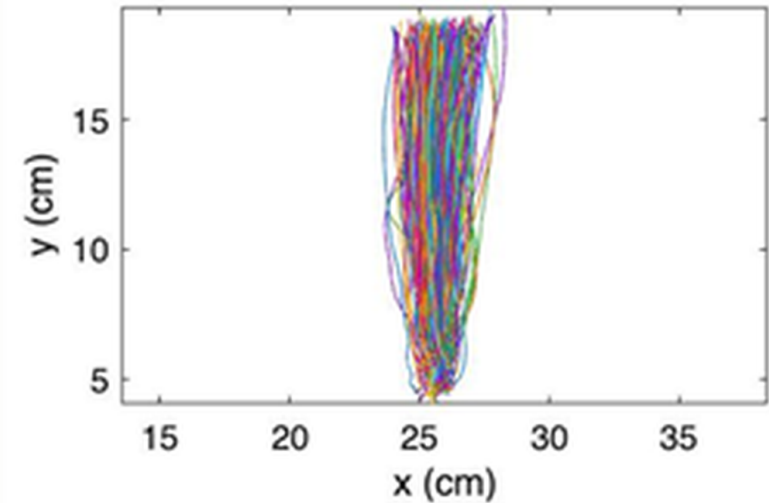
THE QUESTION

Everyone moves differently. Can we compress that?

- Slide a finger from a start square to a fixed target square.
Arrive exactly as a moving circle passes through it - under one second.
- 28 participants, 4,732 trials, finger tracked in 3D at 240 Hz.

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numbers per person?



One participant, every trial. Equal x and y scale.

Target is a distribution of movements, not one trajectory - perception and movement are stochastic.

THE IDEAS

How each model compresses a trial

Spline + PCA

$x \rightarrow 18 \text{ spline coefficients} \rightarrow \text{PCA} \rightarrow z$

Fit a smooth curve, keep its coefficients, then keep the n directions that vary most across training trials. Linear throughout.

AE

$x \rightarrow \text{encoder} \rightarrow z \rightarrow \text{decoder} \rightarrow x_{\text{hat}}$

Squeeze a trial through n numbers and rebuild it. Nothing shapes the latent space.

VAE

$x \rightarrow q(z|x) \rightarrow z \rightarrow \text{decoder} \rightarrow x_{\text{hat}}$

The code becomes a distribution. A KL term pulls it toward $N(0, I)$, so the space is continuous and samplable.

CVAE

$x, c \rightarrow q(z|x, c) \rightarrow z, c \rightarrow \text{decoder} \rightarrow x_{\text{hat}}$

The task condition c goes to both halves, so the latent need not spend capacity on task-driven variation.

What we actually minimise

$$L = \text{MSE}(\text{trajectory}) + 20 \cdot \text{MSE}(\text{log durations}) + \beta \cdot \text{KL}(q(z | x, c) \parallel N(0, I))$$

the 100×2 shape

initiation and movement time

β ramps $0 \rightarrow 1$ over 50 epochs

Timing is never an encoder input - it is only a target. The weight 20 was a documented compromise, not tuned.

APPROACH

What we compare

Spline + PCA - linear baseline

Cubic spline, five fixed knots -> 18 coefficients.
PCA on training participants compresses to n .
Timing from a separate Ridge on the scores.
About 10^2 parameters. No learned nonlinearity.

VAE - best neural result

Encoder -> Gaussian latent (n) -> decoder.
Conditions zeroed: it never sees the task.
Decodes shape plus two log-durations.
About 10^5 parameters.

Two ablations - each changes one design choice

Condition-only Ridge - the floor

Task conditions only, no person information.
Answers: how much needs a fingerprint at all?

CVAE

Conditions switched on, for both encoder and decoder.
The model we set out to build.

Conditional AE

Conditions on, but no stochastic bottleneck: the latent is a point.

Latent sizes $n = 2, 3, 4, 8$. Four folds x 17 train / 4 validation / 7 test participants; every participant held out exactly once.

TASK 1 | RECONSTRUCTION

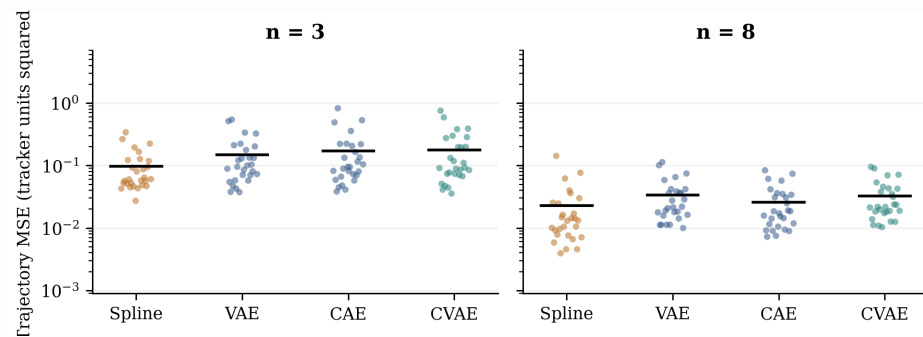
Can it redraw a movement it just saw?

Model	n=3	n=8
Spline + PCA	0.097	0.023
VAE	0.148	0.033
Condition-only Ridge	1.234	1.234

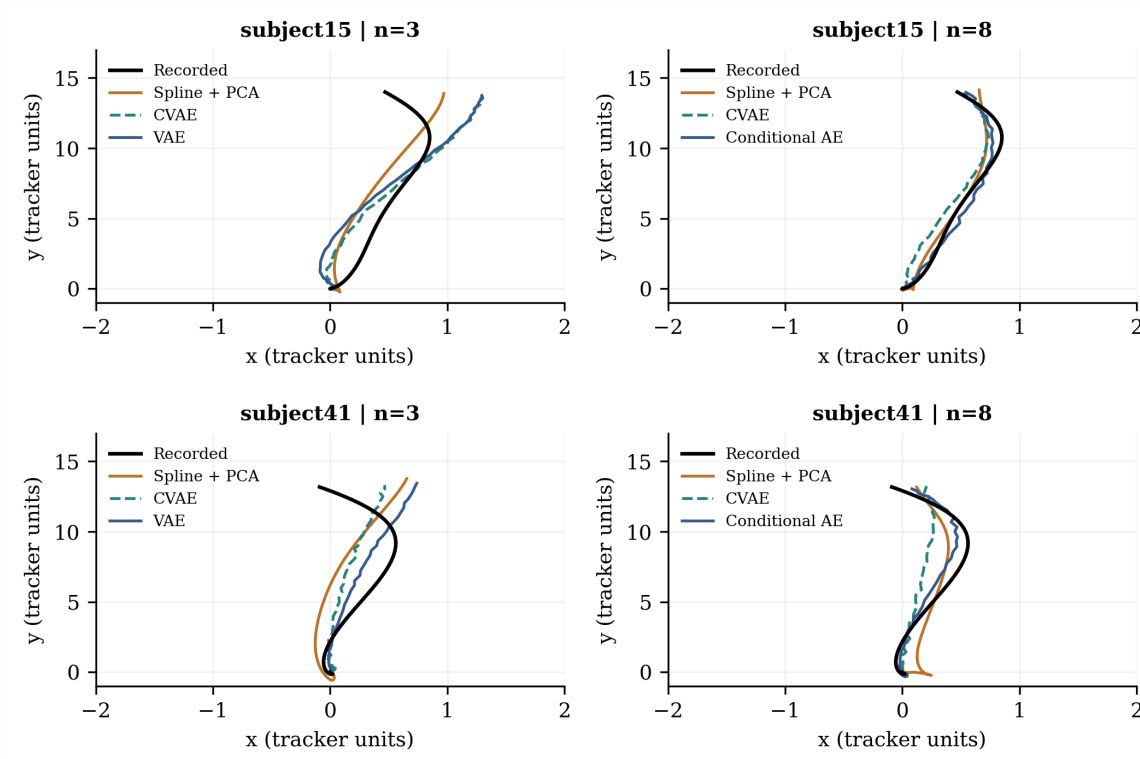
Participant-balanced MSE, tracker units squared. Lower is better.

- The linear baseline wins.

PCA is optimal for squared error; the VAE trades accuracy for a samplable space.



Per participant, log axis.



Recorded input vs decoded output, seed 42. Note: VAE at n=3, CAE at n=8, and the lateral axis is expanded.

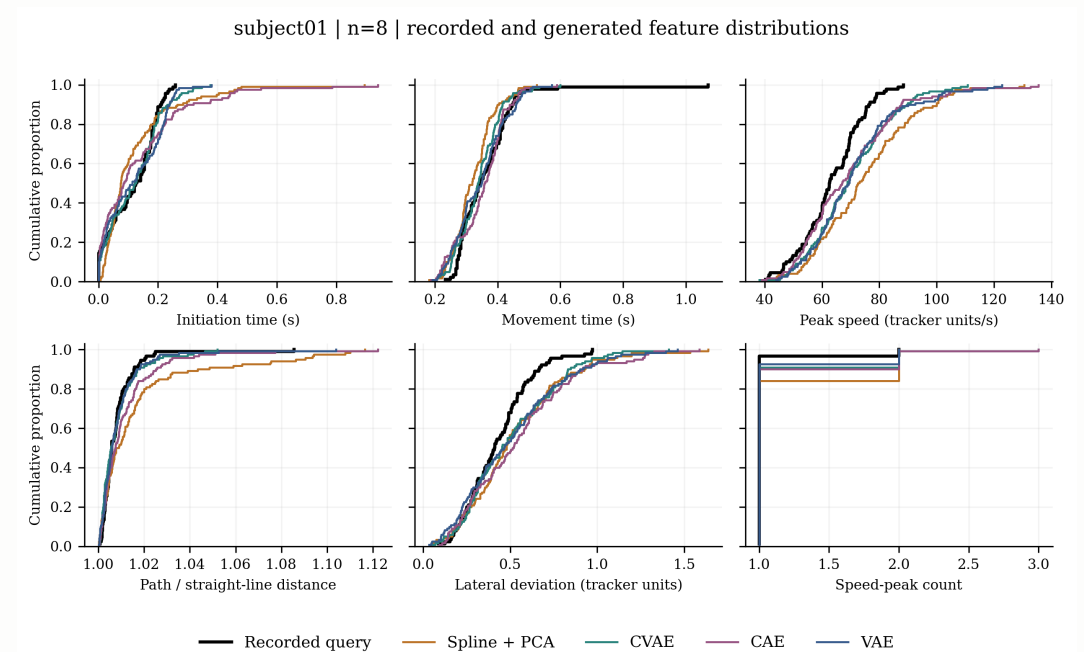
HOW WE MEASURE IT

What does 'matching a person' actually mean?

- Not trajectory-by-trajectory. We compare distributions.
- Every trial becomes 11 kinematic features:

initiation time . movement time . peak speed . time to peak
path length . straight-line distance . curvature . lateral deviation
speed-peak count . endpoint x . endpoint y

- Then: KS per feature, plus energy distance
and MMD across all 11 jointly.



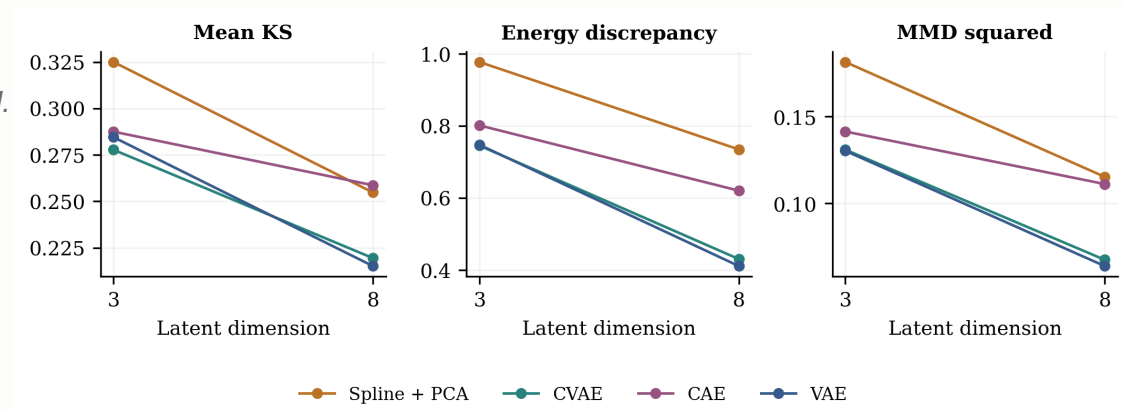
TASK 2 | GENERATION

Can it invent new movements for a person?

- Give the model ~85 of a person's trials. Never show it the rest.
- It generates 120 new trajectories. Compare against the held-out half.

Model (n=8)	Mean KS	Energy	MMD2
VAE	0.215	0.411	0.064
Spline + PCA	0.255	0.735	0.115

The VAE generates closer distributions - and it never sees the task conditions at all.



THE CONTROL THAT MATTERS

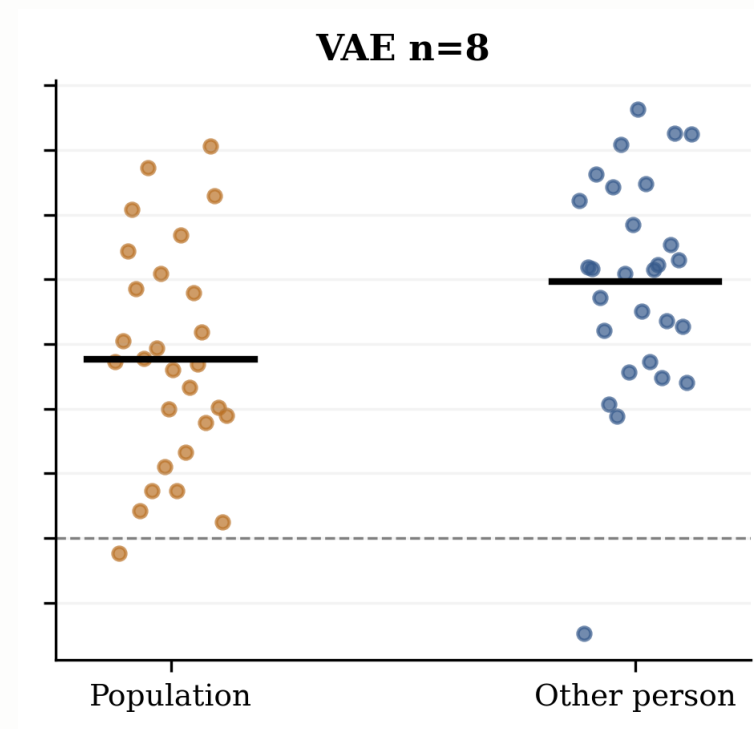
Does the RIGHT person's fingerprint matter?

- Freeze everything: decoder, task draws, noise, covariance, distance reference.
- Change one thing only - whose fingerprint we supply.

VAE n=8, fingerprint supplied	Mean KS	Energy	MMD2
Own participant	0.215	0.411	0.064
Population average	0.284	0.879	0.142
Another participant	0.314	1.168	0.195

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participants improve with their own fingerprint; all 6 contrasts $p < 1.5e-06$



HONEST SUMMARY

What holds, and what does not

Holds

- Personal information really is in the fingerprint - under a strict control.
- The VAE generates closer feature distributions than the baseline.

Does not

- Spline + PCA reconstructs better.
- The best generator ignores the task entirely.
- A simple average beats it on all 14 targets.

If you are building this today:

reconstruct with Spline + PCA; generate with the VAE. Use $n=8$ - $n=3$ is worse on almost every endpoint.

Caveat: n was matched for comparability, not chosen by a variance or elbow criterion. That criterion is the next step.

DEMO

See it work

- Pick a model and capacity, pick a participant.
- Move the latent sliders - watch the trajectory change.
- Compare generated against that person's real held-out trials.

